

Executive Summary

Project Overview

Business Problems

- Identify the cities and markets contributing the largest share of Airbnb listings.
- Understand how listing growth and market presence vary across locations and time.
- Compare average pricing across different accommodation types.
- Evaluate guest satisfaction across key rating dimensions.
- Understand reviewer engagement and the prevalence of repeat reviewers.
- Assess host trust through identity verification and profile-picture adoption.
- Identify seasonal variations in review activity across major markets.

Business Questions

1. Which cities contribute the largest share of Airbnb's global listings?
2. How concentrated is Airbnb's listing portfolio across major markets?
3. How does average price vary by accommodation type?
4. How do cities compare in terms of overall guest ratings?
5. Which rating dimensions—Accuracy, Check-in, Cleanliness, Communication, Location, and Value—perform strongest or weakest?
6. What proportion of reviewers contribute one review versus multiple reviews?
7. How significant is repeat-reviewer behavior?
8. What proportion of hosts are identity-verified and have a profile picture?
9. How does review activity vary across major cities throughout the year?
10. What seasonal patterns can be identified across key Airbnb markets?

Key Insights

- **Market concentration:** The top 5 cities—Paris, New York, Sydney, Rome, and Rio de Janeiro—account for approximately **68% of total listings**, while the top 7 cities contribute around **84%**, demonstrating a highly concentrated marketplace.
- **Pricing differentiation:** **Hotel rooms have the highest average price at approximately \$800**, followed by Entire Places at \$673, Shared Rooms at \$580, and Private Rooms at \$462.
- **Strong guest satisfaction:** Most cities maintain ratings of **9.2+ across the major rating dimensions**, with Check-in and Communication consistently among the strongest-performing factors.
- **Limited repeat-review behavior:** **86.5% of reviewers have posted only one review**, while **96.6% have posted two or fewer reviews**. The cumulative share

reaches approximately **99.9% by five reviews**, highlighting the dominance of one-time reviewers.

- **Strong trust signal: 66.9% of hosts are both identity-verified and have a profile picture**, making this the dominant host-trust segment.
- **Seasonal market behavior:** Review activity demonstrates distinct seasonal patterns across cities, with **Paris and Rome showing stronger representation around June–August, New York gaining prominence during October–November, and Sydney showing stronger activity toward November–December.**

Business Impact

The analysis provides a consolidated view of Airbnb’s marketplace performance and helps stakeholders identify **high-contribution markets, pricing opportunities, customer satisfaction trends, reviewer engagement patterns, trust signals, and seasonal demand behavior.**

These insights can support decisions related to **market prioritization, pricing strategy, host quality initiatives, customer experience improvement, and seasonal planning.**

Tools & Techniques

- **Power BI**
- **DAX**
- Interactive KPI Cards
- Combo Charts
- Pareto / Cumulative Analysis
- Rating Heatmaps
- Ribbon Charts
- Dynamic Filtering & Drill-through
- Conditional Formatting
- Data Segmentation
- Time-Series & Seasonality Analysis
- Interactive Dashboard Design